The role of verbal working memory in predictability effects during reading

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Readers can use context to predict upcoming information, yielding well-established predictability effects in reading [1,2]. However, it is unclear whether these effects reflect resource-intensive predictive mechanisms or resource-free pre-activation. Theories of prediction agree that prediction engages both a resource-free mechanism driven by lexical associations and a resource-intensive mechanism requiring working memory (WM) resources [3,4]. However, the use of the resource-intensive mechanism may be optional and contingent on readers' WM resources. To test this possibility, this study examined whether predictability effects in eye-tracking during reading are modulated by the availability of verbal WM resources and by individual differences in verbal WM capacity.

Native English speakers (N = 81) read 88 sentences in which the target word (e.g., *ink*) was either predictable (e.g., *His green pen had just run out of ink in the office*. M cloze = .95) or unpredictable (e.g., *The supply cabinet had just run out of ink in the office*. M cloze = 0). Participants answered a comprehension question on every trial (M accuracy = 92%). Before reading each sentence, they memorised five words (high-load condition) or two words (low-load condition) that were unrelated to the sentence for later recall (**Fig. 1**). If a resource-intensive mechanism drives predictability effects during reading, these effects should be reduced in the high-load condition relative to the low-load condition [5,6]. Additionally, this interaction should be more pronounced in participants with lower verbal WM capacity (indexed by verbal recall performance) because they should be more susceptible to WM load.

Mean **recall accuracy** (proportion of correctly recalled words) was higher in the low-load (93%, SD = 6) than in the high-load condition (78%, SD = 13), suggesting that the WM task was more difficult under high load. The **eye-tracking results** replicated prior studies [1,2] and showed shorter gaze duration, $\beta = -0.03$, $SE = 0.006$, $t = -5.7$, and higher skipping rate, $\beta = 0.09$, $SE = 0.03$, $z = 2.9$, $p = .004$, for high-cloze versus low-cloze targets (**Fig. 2**). The skipping rate was also higher in the high-load than in the low-load condition, $\beta = 0.16$, $SE = 0.04$, $z = 4.4$, $p < .001$. However, predictability effects did not interact with load (gaze duration: $\beta = 0.0007$, $SE = 0.005$, $t = 0.1$, skipping rate: $\beta = 0.006$, $SE = 0.03$, $z = 0.2$, $p = .8$). The **individual-differences analysis on gaze duration** showed that participants with lower recall accuracy had longer gaze durations than those with higher recall accuracy, $\beta = -0.04$, $SE = 0.01$, $t = -2.8$ (**Fig. 3A**), with no interaction. In contrast, the **individual differences analysis on skipping rate** revealed a main effect of load, $\beta = 0.16$, $SE = 0.03$, $z = 6.1$, $p < .001$, a predictability × recall accuracy interaction, $\beta = -0.06$, $SE = 0.03$, $z = -2.2$, $p = .03$, and a load × recall accuracy interaction, $\beta = 0.08$, $SE = 0.03$, $z = 2.8$, $p = .004$ (**Fig. 3B**). No other interactions were significant.

Taken together, the predictability effect on **gaze duration** was robust across load conditions and participants, suggesting that facilitation of reading time by predictability is largely driven by a resource-free mechanism in native language comprehension. By contrast, the predictability effect on **skipping rate** was larger in participants with *lower* recall accuracy. This finding is inconsistent with both the resource-intensive hypothesis (which predicts a larger predictability effect in participants with *higher* recall accuracy) and the resource-free hypothesis (which predicts no interaction). Participants with higher recall accuracy also showed a greater increase in skipping under high versus low load than those with lower accuracy. Thus, they may have strategically allocated resources between the recall and reading tasks by skipping more words under high load, thereby attenuating the predictability effect on skipping. These findings suggest that predictability effects on skipping are more subject to strategic control than effects on gaze duration. Facilitation of reading speed and skipping may be supported by mechanisms that differ in the extent to which they involve resource-intensive processes.

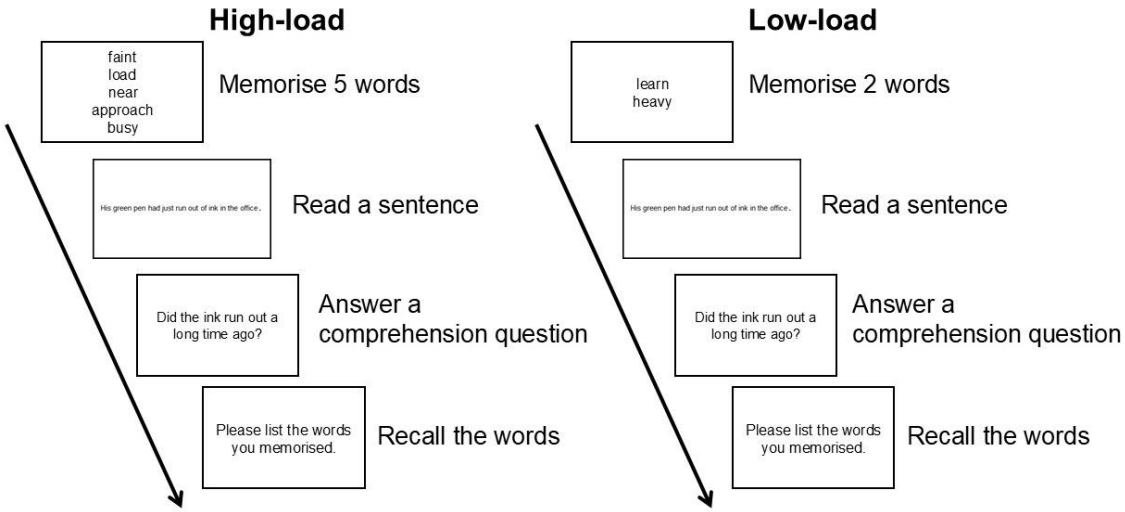

Fig. 1. An illustration of the task.

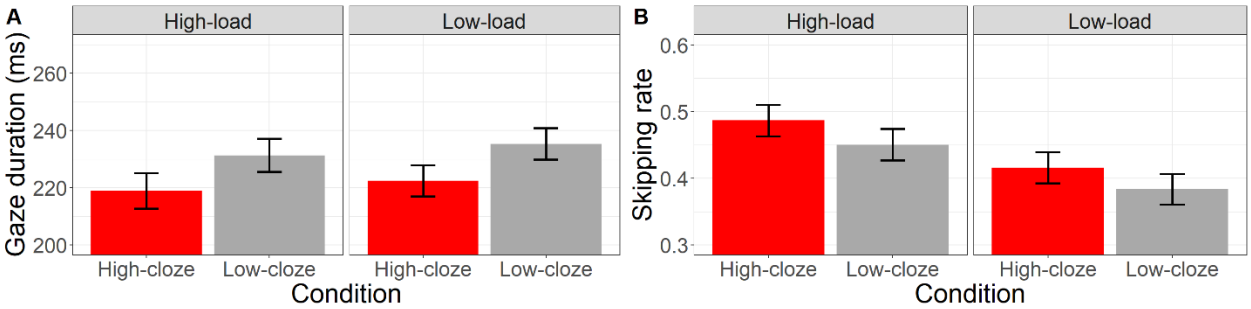

Fig. 2. (A) Gaze duration and **(B)** skipping rate for the target word. Error bars indicate 95% confidence intervals.

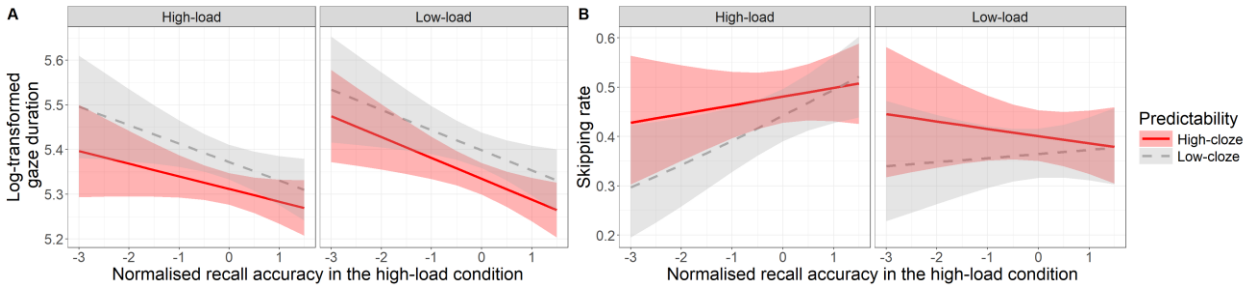

Fig. 3. (A) The estimated marginal means from the model testing a 3-way interaction of predictability by load by recall accuracy on **gaze duration**. **(B)** The estimated marginal means from the model testing a 3-way interaction of predictability by load by recall accuracy on **skipping rate**.
Note. Recall accuracy was taken from the high-load condition because data in the low-load condition showed a ceiling effect.

References

[1] Staub (2015). *Lang. Linguist. Compass*. [2] Wong et al. (2024). *Psychon. Bull. Rev*. [3] Pickering & Gambi (2018). *Psychol Bull*. [4] Huettig (2015). *Brain Res*. [5] Ito et al. (2018). *Biling. Lang. Cogn*. [6] Liu et al. (2025). *Sci. Rep*.